SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – APPLICATION BASED QUESTIONS

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO2	**Assessment:**	Application Based Questions, Inclass
Weightage:	_____ 40% of CIA	**Duration / Marks:**	60 mins
CO2 Statement:	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)		
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Section / Batch:	CSE(CS)/2025	**Date:**	21-10-2026

Code of Conduct

I C Veera Bharadwaj Kumar(192565061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read each question carefully before answering (in-Class Assessment)
- Each question is a real-world application scenario. Identify the AI technique that fits, apply it step by step, and justify every decision.
- Questions are drawn from Informed Search (Greedy, A*, heuristics), Local Search and Online Search, Constraint Satisfaction Problems, and Adversarial Search (Minimax, Alpha-Beta, evaluation functions).
- Show intermediate tables (open lists, domains, α/β values) and state the final answer clearly.
- Marks will be awarded using the Application Based Questions rubric: understanding of problem context, application of concepts, problem-solving approach, accuracy of solution and clarity.
- Upload your work in GCR and Github.

Section / Topic	Focus	Question Nos.
1. Informed Search Strategies	Greedy best-first search, A* search, heuristic functions (admissibility, consistency)	**Q1**

2. Local Search, Online Search & Constraint Satisfaction	Local search and optimization problems, online search agent, CSP (MRV, degree heuristic, forward checking)	**Q2, Q3**
3. Game Playing: Adversarial Search	Games, minimax algorithm, alphabeta pruning, evaluation functions	**Q4**

Rubrics for Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Understanding of Problem Context	**20%**	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	**30%**	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	**20%**	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	**20%**	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	**10%**	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (2)	Accuracy of Solution (2)	Clarity (1)	Sub. Total (10)	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A – Application Based Questions

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

Topic: Informed search strategies, heuristic functions | 15 minutes

A campus delivery robot must carry a parcel from the Store (S) to the Hostel (G). The one-way road network and the robot's heuristic estimate h(n) of the remaining travel cost to G (in minutes) are given below.

From	To	Cost
S	A	1
S	B	4
A	C	2
A	D	5
B	G	5
C	D	1
D	G	2

Node	S	A	B	C	D	G
h(n)	6	5	3	3	2	0

Assume ties are broken alphabetically and that a node is re-opened if a cheaper path to it is found.

(a) Apply Greedy Best-First Search from S to G. Write the order in which nodes are expanded, the path returned and its total cost.

(b) Apply A* search. Show the open list with g(n), h(n) and f(n) after each expansion, the order of expansion, and the final path and cost.

(c) Check whether h(n) is admissible and consistent by verifying the required condition for every node and every road.

(d) Compare the outputs of (a) and (b). Explain why Greedy Best-First Search returned a different result and which algorithm you would deploy for the robot, with justification.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

Topic: Local search algorithms and optimization problems, online search agent | 15 minutes

A technician wants to place a Wi-Fi access point at the best position along a corridor with 10 candidate positions x = 0 to 9. The measured signal quality f(x) (higher is better) is:

x	0	1	2	3	4	5	6	7	8	9
f(x)	2	5	9	6	4	7	12	10	8	3

A move shifts the access point to an adjacent position ($x - 1$ or $x + 1$). Steepest-ascent hill climbing moves to the better neighbour and stops when no neighbour is strictly better.

(a) Run steepest-ascent hill climbing from the start positions x = 0, x = 4 and x = 9. Show each step and the final position and signal quality.

(b) Identify all local maxima and the global maximum. If the start position is chosen uniformly at random from the 10 positions, find the probability that one run reaches the global maximum, the probability that at least one of 3 random-restart runs does, and the expected number of runs needed.

(c) A simulated annealing search is at x = 2 and proposes the move to x = 3. Compute the acceptance probability e^(Δ/T) at T = 2 and at T = 1, and explain what this does to the chance of escaping the local maximum as T falls.

(d) The same search is later done by a robot that cannot see the whole corridor, can only measure f(x) where it stands, and cannot undo exploration for free. Explain why this is an online search problem, and state what the agent must remember to avoid cycling.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

Topic: Constraint satisfaction problem (MRV, degree heuristic, forward checking) | 15 minutes

Five exams A, B, C, D and E must each be given one slot from {1, 2, 3}. Two exams that share students must not be in the same slot. The pairs sharing students are: A–B, A–C, A–D, B–C, B–D, C–E, D–E. Because of invigilator availability, exam A can only be in slots 1 or 2, and exam E only in slots 1 or 2.

Exam	Initial domain
A	{1, 2}
B	{1, 2, 3}
C	{1, 2, 3}
D	{1, 2, 3}
E	{1, 2}

Rules for the search: choose the variable using Minimum Remaining Values (MRV), break ties with the degree heuristic, then alphabetically; try values in ascending order; use forward checking after every assignment.

(a) Draw the constraint graph (list its edges) and compute the degree of every exam.

(b) Select the first exam to assign using MRV and the degree heuristic, justify the choice, assign its first value, and show the domains of all exams after forward checking.

(c) Continue the search with the same rules, showing the chosen exam, its value and the updated domains at each step, until a complete assignment is found. Verify it against every constraint.

(d) Interpret: explain how MRV, the degree heuristic and forward checking reduced the search effort here, and what forward checking would signal if a domain became empty.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

Topic: Game playing, adversarial search, evaluation functions | 15 minutes

A game-playing agent (MAX) searches two plies ahead: MAX chooses one of three moves (B, C, D), then MIN replies with one of three moves. Positions at depth 2 are scored by the evaluation function

$$\mathbf{Eval = 2 \times M + 1 \times Mo}$$

where M is the material advantage of MAX and Mo is the mobility advantage of MAX. The leaf features (M, Mo) are given for the leaves in left-to-right order:

MIN node	Leaf 1 (M, Mo)	Leaf 2 (M, Mo)	Leaf 3 (M, Mo)

B	(1, 1)	(5, 2)	(3, 2)
C	(1, 0)	(1, 2)	(2, 2)
D	(6, 2)	(2, 1)	(0, 2)

(a) Compute the Eval value of all nine leaves.

(b) Run plain minimax. Give the backed-up value of B, C and D, the value at the root and the move MAX should play.

(c) Run alpha-beta pruning on the same tree (left to right). Show α and β at each node, mark every pruned branch, and state the root value and the number of leaves evaluated.

(d) If the leaves under C were ordered (6, 4, 2) instead, how many leaves would alpha-beta evaluate? Explain the effect of move ordering and why pruning never changes the minimax value.

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

(a) Greedy Best-First Search

Greedy Best-First Search selects the node with the lowest heuristic value h(n), without considering the cost already travelled. Its evaluation function is f(n) = h(n).

Step	Expanded node	Open list after expansion	Closed list
1	S (h=6)	A:5, B:3	S
2	B (h=3)	A:5, G:0	S, B
3	G (h=0) – goal	Empty	S, B, G

Expansion order: S → B → G.

Path returned: S → B → G.

Total cost = cost(S→B) + cost(B→G) = 4 + 5 = 9 minutes.

(b) A* Search

A* uses f(n) = g(n) + h(n), where g(n) is the actual cost from the start and h(n) is the estimated remaining cost.

Step	Expanded node	Open list after expansion (g,h,f)	Remarks
0	—	S (0,6,6)	Start
1	S (0,6,6)	A (1,5,6); B (4,3,7)	A and B generated
2	A (1,5,6)	C (3,3,6); B (4,3,7); D (6,2,8)	C and D generated
3	C (3,3,6)	D (4,2,6); B (4,3,7)	D improved from g=6 to g=4
4	D (4,2,6)	G (6,0,6); B (4,3,7)	G generated

5	G (6,0,6)	B (4,3,7) remains	Goal selected; stop

Expansion order: S → A → C → D → G.

Final path: S → A → C → D → G.

Total cost = 1 + 2 + 1 + 2 = 6 minutes. This is the optimal path.

(c) Admissibility and consistency

A heuristic is admissible when $h(n) \leq h^*(n)$, where $h^*(n)$ is the true cheapest cost to the goal.

Node	Cheapest path to G	h*(n)	h(n)	Admissible?
S	S-A-C-D-G	6	6	Yes
A	A-C-D-G	5	5	Yes
B	B-G	5	3	Yes
C	C-D-G	3	3	Yes
D	D-G	2	2	Yes
G	G	0	0	Yes

A heuristic is consistent when $h(n) \leq c(n,n') + h(n')$ for every directed edge.

Edge	h(n)	c(n,n′) + h(n′)	Consistent?
S→A	6	1+5=6	Yes
S→B	6	4+3=7	Yes
A→C	5	2+3=5	Yes
A→D	5	5+2=7	Yes
B→G	3	5+0=5	Yes

C→D	3	1+2=3	Yes
D→G	2	2+0=2	Yes

Therefore, the heuristic is both admissible and consistent.

(d) Comparison and recommendation

Feature	Greedy Best-First Search	A* Search
Evaluation	f(n)=h(n)	f(n)=g(n)+h(n)
Expansion order	S, B, G	S, A, C, D, G
Path	S-B-G	S-A-C-D-G
Cost	9 minutes	6 minutes
Optimality	Not guaranteed	Optimal under the given conditions

Greedy chooses B because h(B)=3 is less than h(A)=5, but it ignores the cost already travelled and returns a path costing 9 minutes. A* considers both travelled and estimated remaining costs and finds the 6-minute route. A* is recommended because it finds the least-cost route for this problem. Greedy is not guaranteed to find an optimal path.

QUESTION 2: WI-FI ACCESS POINT PLACEMENT – LOCAL SEARCH AND ONLINE SEARCH

(a) Steepest-ascent hill climbing

Steepest-ascent hill climbing moves to the better adjacent position and stops when no neighbour is strictly better.

x	0	1	2	3	4	5	6	7	8	9
f(x)	2	5	9	6	4	7	12	10	8	3

Start at x=0

Current x	f(x)	Neighbour values	Decision
0	2	f(1)=5	Move to 1
1	5	f(0)=2, f(2)=9	Move to 2
2	9	f(1)=5, f(3)=6	Stop

Final position: x=2; signal quality f(2)=9. This is a local maximum.

Start at x=4

Current x	f(x)	Neighbour values	Decision
4	4	f(3)=6, f(5)=7	Move to 5
5	7	f(4)=4, f(6)=12	Move to 6
6	12	f(5)=7, f(7)=10	Stop

Final position: x=6; signal quality f(6)=12. This is the global maximum.

Start at x=9

Current x	f(x)	Neighbour values	Decision
9	3	f(8)=8	Move to 8
8	8	f(7)=10, f(9)=3	Move to 7
7	10	f(6)=12, f(8)=8	Move to 6
6	12	f(5)=7, f(7)=10	Stop

Final position: x=6; signal quality f(6)=12. This is the global maximum.

(b) Local/global maxima and probabilities

Local maxima: x=2 with f(2)=9, and x=6 with f(6)=12. Global maximum: x=6, f(x)=12.

Starting position	0	1	2	3	4	5	6	7	8	9
Final position	2	2	2	2	6	6	6	6	6	6

Six of the ten starting positions reach the global maximum.

Probability of success in one run: p = 6/10 = 0.6 = 60%.

For three independent random-restart runs, probability of at least one success = 1 − (1−p)^3 = 1 − (0.4)^3 = 1 − 0.064 = 0.936 = 93.6%.

Expected number of independent runs until the first success = 1/p = 1/0.6 ≈ 1.67 runs.

(c) Simulated annealing acceptance probability

Current position x=2 has f(2)=9. Proposed position x=3 has f(3)=6.

Δ = f(new) − f(current) = 6 − 9 = −3. Acceptance probability = e^(Δ/T).

Temperature	Calculation	Acceptance probability
T=2	e^(−3/2) = e^(−1.5)	0.2231 ≈ 22.31%
T=1	e^(−3)	0.0498 ≈ 4.98%

At higher temperatures, the algorithm is more likely to accept worse moves and escape local maxima. As temperature decreases, the chance of accepting worse moves falls, making escape less likely.

(d) Online search

This is an online search problem because the robot cannot see the entire corridor in advance. It can measure f(x) only at its current position, and each movement is a real action. The agent must alternate between acting and learning about the environment.

To avoid cycling, it should remember: (1) visited positions and measured signal values; (2) actions already tried at each position; (3) neighbouring positions and action outcomes; (4) predecessor states or a backtracking record; and (5) the best position and signal value found so far. Without this information, the robot could repeatedly move between positions.

QUESTION 3: EXAM TIMETABLE SCHEDULING – CONSTRAINT SATISFACTION PROBLEM

(a) Constraint graph and degrees

Variables are exams A, B, C, D and E. A and E have domains {1,2}; B, C and D have domains {1,2,3}. Connected exams must use different time slots.

Edges: A–B, A–C, A–D, B–C, B–D, C–E and D–E.

Exam	Neighbours	Degree
A	B, C, D	3
B	A, C, D	3
C	A, B, E	3
D	A, B, E	3
E	C, D	2

(b) First variable selection using MRV and degree heuristic

Exam	Initial domain	Domain size	Degree
A	{1,2}	2	3
B	{1,2,3}	3	3
C	{1,2,3}	3	3
D	{1,2,3}	3	3
E	{1,2}	2	2

MRV selects A or E because both have two values. The degree heuristic breaks the tie: degree(A)=3 is greater than degree(E)=2, so choose A. Try the lowest value, A=1. Forward checking removes 1 from the domains of B, C and D.

Exam	Domain after A=1
A	{1}
B	{2,3}
C	{2,3}
D	{2,3}
E	{1,2}

(c) Continue the search

Step	Selected exam	Value	Domains after forward checking (A, B, C, D, E)
1	A	1	A={1}, B={2,3}, C={2,3}, D={2,3}, E={1,2}
2	B	2	A={1}, B={2}, C={3}, D={3}, E={1,2}
3	C	3	A={1}, B={2}, C={3}, D={3}, E={1,2}
4	D	3	A={1}, B={2}, C={3}, D={3}, E={1,2}
5	E	1	A={1}, B={2}, C={3}, D={3}, E={1}

Complete assignment: A=1, B=2, C=3, D=3, E=1.

Constraint	Values	Satisfied?
A≠B	1≠2	Yes
A≠C	1≠3	Yes
A≠D	1≠3	Yes
B≠C	2≠3	Yes
B≠D	2≠3	Yes

C≠E	3≠1	Yes
D≠E	3≠1	Yes
A and E in {1,2}	A=1, E=1	Yes

C and D can both use slot 3 because there is no constraint between C and D.

(d) Interpretation

MRV selects the most constrained variable first, helping detect failure early. The degree heuristic breaks ties by choosing the variable that constrains the largest number of other variables. Forward checking removes values that conflict with each assignment. After B=2, C and D are forced to {3}, while E remains {1,2} because E is not connected to B. If any domain becomes empty, the current partial assignment cannot lead to a valid solution; the algorithm must backtrack and try another value.

QUESTION 4: BOARD-GAME AI – MINIMAX, ALPHA-BETA PRUNING AND EVALUATION FUNCTION

(a) Leaf evaluation values

Evaluation function: Eval = 2 × M + 1 × Mo, where M is the material advantage of MAX and Mo is the mobility advantage of MAX.

MIN node	Leaf 1	Leaf 2	Leaf 3
B	(1,1): 2×1+1=3	(5,2): 2×5+2=12	(3,2): 2×3+2=8
C	(1,0): 2×1+0=2	(1,2): 2×1+2=4	(2,2): 2×2+2=6
D	(6,2): 2×6+2=14	(2,1): 2×2+1=5	(0,2): 2×0+2=2

(b) Plain minimax

B is a MIN node: B = min(3,12,8) = 3.

C is a MIN node: C = min(2,4,6) = 2.

D is a MIN node: D = min(14,5,2) = 2.

The root is a MAX node: Root = max(3,2,2) = 3.

Final answer: root value = 3; MAX should play move B. Plain minimax evaluates all 9 leaves.

(c) Alpha-beta pruning, left to right

Initially $\alpha = -\infty$ and $\beta = +\infty$. Prune when $\beta \leq \alpha$.

At B, evaluate 3, 12 and 8. B returns 3. At the root, α becomes 3.

At C, the first leaf is 2, so β becomes 2. Since $\beta=2 \leq \alpha=3$, prune the remaining leaves under C (values 4 and 6). C returns 2.

At D, evaluate 14, then 5, then 2. The cutoff condition $\beta \leq \alpha$ is reached only after the last leaf, so no leaves remain to be skipped. D returns 2.

Node	Leaves evaluated	Returned value	Pruning
B	3, 12, 8	3	None
C	2	2	Prune leaves 4 and 6
D	14, 5, 2	2	Cutoff at last leaf; none skipped

Total leaves evaluated = 3 + 1 + 3 = 7. Leaves pruned = 9 − 7 = 2. Root value = 3, so MAX selects B.

(d) Effect of move ordering

If C's leaves are ordered (6,4,2), with $\alpha=3$ at the root, first evaluate 6: $\beta=6$, so continue. Then evaluate 4: $\beta=4$, so continue. Finally evaluate 2: $\beta=2 \leq \alpha=3$, so the cutoff occurs only after the last leaf under C. No leaf is pruned under C.

D also evaluates all three leaves because its cutoff occurs at its final leaf. Total leaves evaluated = 3 + 3 + 3 = 9.

Move ordering affects efficiency. Examining promising moves first can produce earlier cutoffs and more pruning; poor ordering can reduce or eliminate the benefit. Alpha-beta pruning does not change the minimax value because it removes only branches that cannot affect the final decision. The root value remains 3 and MAX still selects B.

CONCLUSION

These questions demonstrate informed search, local and online search, constraint satisfaction, minimax and alpha-beta pruning. A* finds the least-cost delivery route, hill climbing identifies local and global maxima, CSP heuristics produce a valid exam timetable, and alpha-beta pruning reduces game-tree evaluations without changing the minimax result.